Geo-ΔSSM: Geometry-Conditioned Δ-Modulated Selective SSM Network for Hyperspectral and LiDAR Joint Classification

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***Abstract*—Hyperspectral imagery (HSI) and LiDAR-derived elevation data possess distinct and complementary characteristics, making their joint classification essential for precise land-cover interpretation. However, the intrinsic heterogeneity between high-dimensional spectral signatures and geometric structural priors hinders efficient cross-modal interaction. This phenomenon further exacerbates the difficulty in resolving boundary ambiguities, ultimately resulting in suboptimal performance in geometrically complex regions. In this article, a geometry-conditioned Δ-modulated selective state space model network (Geo-ΔSSM) is proposed to solve these challenges. Our core motivation is to explicitly regulate the spectral modeling dynamics of the HSI branch using LiDAR-derived geometric priors at the parameter level. Unlike conventional feature-level fusion, Geo-ΔSSM injects LiDAR-derived geometry into the generation of the selective SSM parameters, including Δ, B, and C, constraining spectral propagation by object boundaries. To further address spectral redundancy and noise, a frequency enhancement module (FrEM) is constructed to fuse static spectral priors with dynamic geometric guidance in the Fourier domain. The whole structure utilizes a weighted dual-modal boundary-aware loss (DBAL) to penalize errors at both material transitions and elevation transitions. Experiments demonstrate that Geo-ΔSSM achieves superior performance on three public benchmarks, effectively balancing accuracy on minority classes and preserving fine-grained details. The datasets and codes are available at https://github.com/bourne159/Geo-SSM**

***Index Terms*—Boundary-aware learning, geometry-conditioned discretization, hyperspectral, joint classification, light detection and ranging, state space model (SSM).**

I. Introduction

LAND-COVER classification is a fundamental task in remote sensing, supporting various applications such as urban planning, environmental monitoring, and disaster response [1], [2]. With the rapid growth of Earth observation, remote sensing interpretation has increasingly shifted from single-sensor analysis to multimodal learning, in which complementary modalities are jointly exploited to improve robustness and generalization in complex scenes [3]. Among various multimodal combinations, hyperspectral imagery (HSI) and LiDAR-derived digital surface models (DSMs) are particularly complementary: HSI provides dense spectral signatures for fine-grained material discrimination, while DSMs provide explicit elevation and structural features that are less affected by spectral ambiguities and illumination fluctuations. The effectiveness of HSI-LiDAR joint classification has been validated across public benchmarks [4], [5].

Effective cross-modal interaction between HSI and LiDAR data remains challenging in real-world scenes. First, the two modalities are highly heterogeneous. HSI is characterized by high dimensionality, spectral redundancy, and noise, whereas LiDAR data are low-dimensional and mainly encode geometric structure. This disparity renders shallow fusion insufficient for exploiting complementarity and may amplify modality-specific artifacts [5], [6]. Second, boundary ambiguity frequently occurs because HSI material transitions (e.g., asphalt and soil) and LiDAR geometric discontinuities (e.g., building edges and elevation jumps) do not always coincide spatially. Thus, models optimized with uniform pixel-wise objectives tend to produce over-smoothed prediction maps and boundary misclassifications around object borders [7], [8]. Third, long-range dependency modeling is required to enforce global consistency and handle spectrally similar categories, but transformer-style attention introduces quadratic complexity with respect to token length, causing considerable memory overhead in high-resolution patches and dense networks [2].

Early deep fusion methods employed dual-branch convolutional neural networks (CNNs) to comprehensively extract HSI spectral–spatial features and LiDAR structural features, followed by feature-level fusion [9]. Lu et al. [10] further proposed a coupled CNN to encourage cross-modal interaction through shared representations, making it a strong baseline for joint classification. However, CNN-based fusion is constrained by local receptive fields and may require deeper networks and more computation to capture long-range dependencies.

Graph-based fusion provides another paradigm by modeling nonlocal dependencies and structural priors through graph reasoning. Hybrid CNN-graph convolutional network

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(GCN) architectures construct unified graphs to capture mid- and long-range structural dependencies while learning modality-specific HSI and LiDAR representations [11]. Recent graph-attention fusion networks further build heterogeneous token graphs from multisource embeddings, enabling attentive message passing to explore cross-modal semantic relationships beyond simple feature concatenation [12]. Graph reasoning has also been integrated with attention-based mechanisms. For example, multihead self-attention can be augmented with graph convolution to strengthen spatial relational modeling and multimodal feature interaction [13]. Moreover, modality semantics can be aligned through graph-based aggregation strategies [14]. However, graph-driven interaction highly depends on graph construction and may introduce additional modeling complexity and hyperparameter sensitivity, especially for dense prediction on high-resolution patches.

Transformer-based fusion has garnered significant attention for its capacity to model global interactions. Multimodal fusion transformer (MFT) adopts an attention-based integration mechanism and achieves robust performance on cross-modal classification benchmarks [15]. Similarly, multimodal cross-layer fusion transformer network (MCFTNet) combines local convolutional feature extraction with global transformer-based reasoning for joint classification [16]. Despite their effectiveness, these attention-driven architectures suffer from substantial computational and memory overhead, and boundary ambiguities may remain unresolved without explicit boundary-aware constraints or edge-preserving mechanisms.

Multiple approaches focus on improving robustness against spectral redundancy and noise. LiDAR-guided cross-attention is explored for hyperspectral band selection, demonstrating that LiDAR data can guide redundancy reduction and strengthen fusion. Frequency-domain learning suppresses noise and enhances structural textures. Wang et al. [17] proposed a frequency-domain network (FDNet), validating the benefit of fast Fourier transform (FFT)-based frequency-local learning [18]. Frequency enhancement is treated as an auxiliary feature transformation and remains confined to feature-level interaction. Therefore, making HSI-LiDAR interaction adaptive to scene geometry, rather than merely exchanging or reweighting features, presents a critical challenge.

Recently, selective state space models (SSMs) have emerged as efficient alternatives to transformers for long-sequence modeling. Mamba models demonstrate that selective SSMs can achieve linear-time sequence modeling while retaining strong modeling capability [19]. This efficiency is particularly attractive for remote sensing, where dense patches and multi-branch fusion are computationally expensive under quadratic attention. However, directly adopting SSMs in joint classification does not automatically solve the cross-modal interaction challenge. The challenge lies in guiding spectral modulation of the HSI branch using LiDAR geometry, particularly in boundary-rich and geometrically complex regions.

In standard selective SSMs, the selective parameters are mainly generated from the input sequence itself. For HSI data, such input-dependent generation may be affected by redundant bands, mixed spectral responses, and boundary noise, causing spectral information to propagate through geometrically inconsistent regions. Since LiDAR-derived DSMs explicitly record elevation discontinuities and structural boundaries, it is reasonable to use LiDAR geometry to regulate the internal SSM selective parameters. Therefore, our method uses LiDAR-derived priors to guide not only the discretization step but also the input and output mappings of the selective scan.

From a methodological perspective, geometry-conditioned Δ-modulated selective SSM network (Geo-ΔSSM) shifts HSI-LiDAR interaction from feature-level fusion to geometry-conditioned selective SSM parameter generation. Unlike CNN-, transformer-, and Mamba-based fusion methods that typically incorporate LiDAR information through concatenation, attention, or gating, Geo-ΔSSM uses LiDAR-derived geometric priors to generate the selective SSM parameters, including the discretization step, input injection, and state readout. LiDAR geometry directly regulates the internal spectral scanning dynamics of the HSI branch, rather than acting merely as an auxiliary feature. This design constrains spectral propagation with scene geometry while preserving the linear modeling efficiency of selective SSMs. In this article, a geometry-conditioned Δ-modulated selective SSM network called Geo-ΔSSM is presented for efficient HSI and LiDAR joint classification. Our core motivation is to explicitly regulate the spectral modeling dynamics of the HSI branch with assistance from LiDAR-derived geometric priors at the parameter level. Geo-ΔSSM aims to address these challenges, including heterogeneity, boundary ambiguity, and efficiency bottlenecks. The main contributions are summarized as follows.

1) Geo-ΔSSM presents a novel geometry-guided selective parameter modulation (GSPM) strategy, incorporating LiDAR-derived geometric priors into the generation of the selective SSM parameters in the HSI branch. This strategy enables the adaptive adjustment of Δ, B, and C according to structural variations, enabling geometry to explicitly regulate the spectral scanning dynamics. Thus, Geo-ΔSSM enhances cross-modal interaction while preserving the efficient sequence modeling capability of selective SSMs.
2) Geo-ΔSSM develops a dual-guided interaction mechanism, comprising an FFT-based enhancement module that suppresses noise while preserving high-frequency structural details. In addition, geometry-conditioned signals guide the enhancement process to preserve structural textures. Integrated with GSPM, Geo-ΔSSM adaptively responds to both spectral noise and geometric complexity.
3) Geo-ΔSSM designs a weighted dual-modal boundary-aware loss (DBAL), penalizing boundary pixels by leveraging LiDAR gradients for geometric boundaries and HSI gradients for material boundaries. This boundary-focused supervision strategy complements the backbone and enhances prediction fidelity near edge regions. Although boundary-aware learning has proven effective in segmentation tasks, Geo-ΔSSM tailors it for multimodal joint classification. Class-specific weights penalize errors on underrepresented samples, enabling

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better performance when identifying rare objects despite data imbalance.

Section II provides some related works. Section III describes Geo-ΔSSM in detail. Section IV presents experimental results and ablation studies. Section V concludes this article.

II. Related Works

A. HSI and LiDAR Joint Classification

HSI-LiDAR joint classification has evolved from simple feature stacking to interactive fusion schemes for mitigating modality heterogeneity. Although CNN-based methods improve local feature extraction, their limited ability to capture long-range dependencies has encouraged the use of attention-based architectures. Wang et al. [20] designed a multiscale attention fusion network with an improved transformer to capture complementary information across hierarchical levels, and He et al. [21] proposed a multilevel attention dynamic-scale network to adapt receptive fields across sensors. Ge et al. [22] introduced a cross-attention convolutional fusion network to bridge the semantic gap between spectral and elevation features, while Zhang et al. [23] combined transformer- and scale-based methods for complex category discrimination. However, self-attention still suffers from quadratic computational complexity in high-resolution prediction tasks. These methods enhance HSI-LiDAR interaction at the feature level through attention-based or scale-adaptive fusion strategies. However, the interaction is still primarily performed on fused features, while the internal sequence modeling parameters are not explicitly regulated by LiDAR-derived geometric priors.

B. Selective SSMs in Remote Sensing

Recently, selective SSMs, particularly Mamba, have emerged as linear-time alternatives to transformers for long-sequence modeling. A continuous linear SSM maps an input signal $x(t)$ to an output $y(t)$ through an N-dimensional hidden state $h(t) \in \mathbb{R}^N$, which is formulated as

$$\begin{cases} h'(t) = Ah(t) + Bx(t) \\ y(t) = Ch(t) \end{cases} \tag{1}$$

where A, B, and C are projection matrices that map the signal into the latent space and back to the observable output. Unlike conventional SSMs with fixed parameters, Mamba introduces an input-dependent selection mechanism for content-aware sequence modeling [19]. This efficiency is paramount for HSI analysis with dense spectral channels. In remote sensing, RSMamba [24] first applied selective scan to large-scale classification, demonstrating its ability to capture global spatial contexts with lower memory overhead than self-attention. Spectral–spatial Mamba [25] further serialized 3-D cubes to model continuous spectral–spatial signatures. By extending this paradigm to other modalities, He et al. [26] demonstrated the effectiveness of Mamba in pan-sharpening, while Chen et al. [27] developed ChangeMamba to utilize spatio-temporal SSMs for change detection in multitemporal imagery.

However, existing Mamba-based remote sensing methods mainly generate selective SSM parameters from the target input features themselves. In multimodal HSI-LiDAR classification, this limits their ability to leverage LiDAR elevation discontinuities for regulating the internal dynamics of the HSI branch. Although feature concatenation or attention-based fusion improves multimodal representations, the discretization step, input injection, and state readout of the selective scan remain weakly constrained by scene geometry. Geo-ΔSSM addresses this limitation by embedding LiDAR-derived structural priors into selective parameter generation, making the SSM dynamics adaptive to geometric boundaries.

C. Frequency-Domain Analysis and Noise Reduction

Spectral redundancy and sensor-induced noise in HSI manifest as high-frequency fluctuations, which inevitably degrade the robustness of spatial-domain convolutions. Shifting features to the frequency domain has proven effective. Wang et al. [17] constructed a pure FDNet, which allows direct processing of spectral signatures in the Fourier space. This approach effectively isolates semantic information from high-frequency noise without the computational burden of spatial operations. Yang et al. [28] introduced an explainable spatial-frequency multiscale transformer with hybrid architectures, which captures global dependencies by interacting features across spatial and frequency domains. Furthermore, to address the heterogeneity of multimodal data, Ni et al. [18] designed a frequency-domain-based network specifically for HSI and LiDAR joint classification, utilizing spectrum-aware filters to separate structural texture from spectral noise. Different from these approaches, which treat frequency transformation as an independent module or rely solely on fixed learnable filters, Geo-ΔSSM introduces a hybrid frequency modulation mechanism. Specifically, LiDAR-derived geometric priors are leveraged to dynamically rescale a learnable spectral weight matrix. This design combines the stability of global static priors, capturing inherent sensor noise patterns, with the adaptability of geometric modulation, preserving structural textures in structurally complex regions.

D. Boundary-Aware Learning and Edge Preservation

In pixel-wise classification, over-smoothing often causes misclassification around material transitions and geometric edges, motivating explicit boundary modeling. Xie et al. [29] proposed an edge-semantics collaborative network (ESCNet) with multiscale fusion and edge-guided decoders to preserve spatial details. Hou et al. [30] introduced a boundary-sensitive network (BSNet), which uses dynamic hybrid gradient convolution to enhance fine-grained boundary sensitivity. Tang et al. [31] further designed a boundary-aware graph reasoning module that utilizes boundary score maps as priors to refine object shapes. Different from these methods that mainly rely on image-level edge features or predicted boundary maps, DBAL explicitly constructs dual-modal boundary weights from HSI material gradients and LiDAR elevation gradients, providing boundary-aware supervision tailored for joint classification.

E. Learning With Imbalanced and Scarce Samples

Remote sensing benchmarks often suffer from severe class imbalance, where dominant categories bias model optimization

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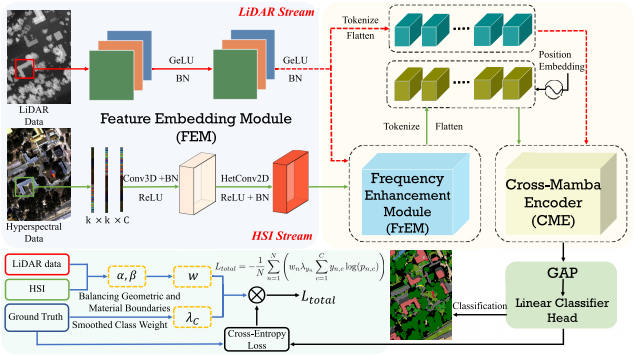


Fig. 1. Scheme of Geo-ΔSSM with three components, i.e., FEM, FrEM, and CME. HSI and LiDAR data-related structures are marked in green and red. Loss-related structures are marked in blue, classification-related structures are marked in black. Solid lines represent standard feature propagation, red dashed lines indicate cross-modal modulation, in which LiDAR-derived geometric priors guide the HSI branch.

and overshadow minority classes. To mitigate this long-tail distribution, Li et al. [32] enhanced minority representations by spectral–spatial feature rotation and ensemble learning. Jiang et al. [33] proposed a dynamic loss framework to reduce majority-class dominance, while Tang et al. [34] used adversarial domain adaptation to alleviate distribution shifts under class imbalance. Different from these methods, mainly focus on global feature enhancement or loss reweighting, Geo-ΔSSM incorporates class-balancing weights into boundary-aware supervision, ensuring that the model maintains high discriminative power for rare classes even in structurally complex transition zones. This design is motivated by the fact that minority classes in HSI-LiDAR benchmarks often appear in small or boundary-rich regions, where class imbalance and boundary ambiguity are closely coupled.

III. Methodology

The architecture of Geo-ΔSSM is illustrated in Fig. 1. The framework comprises three primary components: a feature embedding module (FEM), a frequency enhancement module (FrEM), and a cross-Mamba encoder (CME). In addition, a DBAL is designed to optimize the network by focusing on structural transitions.

A. Preprocessing of HSI and LiDAR Data

The HSI and LiDAR-derived DSM are denoted as $X_H \in \mathbb{R}^{h\times w\times bn}$ and $X_L \in \mathbb{R}^{h\times w}$, respectively, where h, w, and bn denote the height, width, and number of bands. A patch-based processing strategy is adopted. For each pixel (i, j), a spectral–spatial cube patch $x_h^{i,j} \in \mathbb{R}^{k\times k\times bn}$ is segmented from the normalized X_H, and a corresponding elevation cube patch $x_l^{i,j} \in \mathbb{R}^{k\times k}$ is segmented from the normalized X_L, where k is the neighborhood size centered at (i, j).

B. Feature Embedding Module

CNNs have demonstrated robust capabilities in extracting high-level abstractions from HSI spectral bands. Motivated by this capability, a hybrid architecture combining 3-D and 2-D convolutions is considered in Geo-ΔSSM to capture discriminative features. Initially, a 3-D convolution captures local spectral–spatial correlations. Subsequently, heterogeneous kernel-based 2-D (HetConv2D) convolutions [35], which decompose standard convolutions into group-wise and point-wise operations, are utilized to reduce computational redundancy and preserve essential spatial structures. For HSI, the contiguous spectral bands are highly coupled. Thus, 3-D convolutions are essential to capture joint spectral–spatial correlations, while HetConv2D effectively mitigates the parameter explosion typical of 3-D operations. In contrast, LiDAR data are single-channel representations encoding pure physical elevation. Employing overly complex or 3-D operations on LiDAR data is not only redundant but also risks over-smoothing sharp geometric boundaries, e.g., elevation jumps. Standard 2-D convolutions are sufficient and optimal for the LiDAR branch to reliably extract structural priors without geometric distortion. The embedding process is formulated as follows:

$$X^H = Re\left(C_{Het}\left(Re\left(C_{3D}\left(X_{in}^H\right)\right)\right)\right) \tag{2}$$

where $X_{in}^H \in \mathbb{R}^{k\times k\times bn}$ denotes the input HSI patch, X^H is the extracted feature map, $Re(\cdot)$ is the ReLU activation function, and C_{3D} and C_{Het} are the 3-D convolution and HetConv2D. To align the feature dimensions, the LiDAR branch employs two sequential Conv2D layers. The first expands the dimensionality, while the second extracts structural features as follows:

$$X^L = Ge\left(C_{2D}\left(Ge\left(C_{2D}\left(X_{in}^L\right)\right)\right)\right) \tag{3}$$

where the kernel size for both layers is set to 3×3 with a padding of 1, preserving the spatial resolution of $k \times k$. $Ge(\cdot)$ is the GeLU activation function. ReLU is used for the HSI branch to maintain computational efficiency, while GeLU is adopted for the LiDAR branch to preserve smooth gradient flow for structural features. All convolutional layers are followed by batch normalization to mitigate overfitting and accelerate convergence.

C. Frequency Enhancement Module

HSI-related techniques suffer from high-frequency noise, which degrades the performance of downstream models. While Mamba-based architectures work well in sequence modeling, they lack inherent mechanisms for filtering frequency-domain artifacts. To address this limitation, a LiDAR-modulated FrEM is constructed, as shown in Fig. 2. This module integrates static frequency priors with dynamic geometric guidance derived from LiDAR data. Initially, the HSI features are transformed into the frequency domain by FFT. To construct an adaptive filter, a hybrid modulation strategy is considered as follows.

1) *Static Branch:* Initializing a learnable complex weight parameter W_{static} to capture global and dataset-specific noise patterns that are invariant to local texture.
2) *Dynamic Branch:* Generating a content-adaptive scaling factor from the LiDAR feature X^L. Specifically, X^L is processed by global average pooling (GAP), followed by a linear layer and a Sigmoid activation function, to obtain a channel-wise gate γ. This gate reflects the

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structural complexity of the current scene (e.g., boosting high-frequency components in edge-rich areas).

The frequency mask M is obtained by modulating the static weight with the dynamic scaling factor through element-wise multiplication. The enhanced frequency features are transformed back to the spatial domain using inverse FFT (IFFT) and added to the original input by a residual connection. These procedures can be summarized as follows:

$$\gamma = Sig\left(Linear\left(GAP\left(X^L\right)\right)\right) \quad M = W_{static} \odot \gamma$$
$$\hat{X}^H = IFFT\left(M \odot FFT\left(X^H\right)\right) + X^H \tag{4}$$

where $\hat{X}^H, X^L \in \mathbb{R}^{D\times k\times k}$ and D is the feature dimension. FFT and $IFFT$ denote the FFT and IFFT operators, GAP is the GAP operator, $\odot$ represents element-wise multiplication, γ is the dynamic scaling factor derived from LiDAR data, $Sig(\cdot)$ denotes the Sigmoid activation function, and W_{static} is a learnable static spatial-frequency prior defined at the patch level. Before $\odot$, γ is spatially broadcast to match the dimension of W_{static}.

D. Tokenization

To leverage the efficient sequence modeling capabilities of Mamba architectures, the spatial feature maps are explicitly transformed into token sequences in Geo-ΔSSM. This serialization process is essential for capturing long-range dependencies across the spectral–spatial domain, which are often lost in local convolution operations. The enhanced features $\hat{X}^H$ and X^L are projected into token sequences by flattening and transposing their spatial dimensions. This process yields token-based input formats suitable for SSMs as follows:

$$X_{hsi} = T\left(Flatten\left(\hat{X}^H\right)\right) \quad X_{geo} = T\left(Flatten\left(X^L\right)\right) \tag{5}$$

where $X_{hsi}, X_{geo} \in \mathbb{R}^{L\times D}$ represent the tokenized sequences, $L = k \times k$ is the sequence length, and $T(\cdot)$ denotes the matrix transposition operation. Since the flattening operation disrupts the native 2-D spatial structure, the learnable positional embeddings P_{pos} are injected into the HSI sequence to retain spatial context. Notably, as the LiDAR branch primarily serves as an auxiliary guidance for parameter modulation rather than sequential context modeling, the positional embeddings for X_{geo} are omitted to reduce redundancy. The final input token sequence is processed by dropout to prevent overfitting, as $DP(X_{hsi} \oplus P_{pos})$, where $DP(\cdot)$ is the dropout operator (the rate is 0.1), $P_{pos} \in \mathbb{R}^{L\times D}$, and $\oplus$ denotes element-wise addition.

E. Cross-Mamba Encoder

CME is constructed to fuse the HSI spectral semantics with the LiDAR structural priors, which is shown in Fig. 3.

1) Feature Projection and Convolution: The tokenized HSI input $X_{hsi} \in \mathbb{R}^{L\times D}$ is expanded along the feature dimension to create two parallel pathways: the main processing branch data X and the gating branch data Z. This process is achieved by a linear projection followed by a channel split. Before entering the state space equation, one 1-D convolution is applied to X. This step is essential for capturing local spectral–spatial

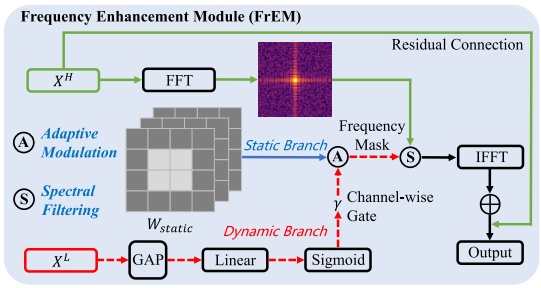


Fig. 2. Detailed architecture of FrEM in Geo-ΔSSM.

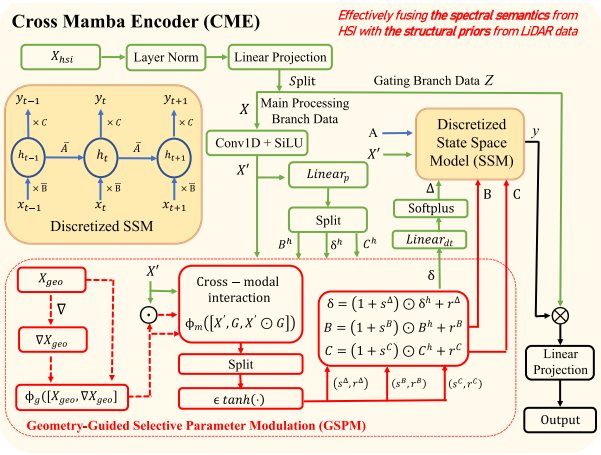


Fig. 3. Detailed architecture of CME in Geo-ΔSSM.

context and preventing the loss of neighborhood information during serialization. These operations are as follows:

$$X, Z = Split\left(Linear\left(X_{hsi}\right)\right) \quad X' = SiLU\left(C_{1D}\left(X\right)\right) \tag{6}$$

where $SiLU(\cdot)$ denotes the SiLU activation function. The expansion factor is set to 2, creating an internal dimension $D_{inner} = 2D$. The gating branch data Z bypasses SSMs to regulate the final information flow.

2) Geometry-Guided Selective Parameter Modulation: The core innovation of Geo-ΔSSM lies in GSPM. In standard Mamba structures, the selective parameters are generated from the input sequence itself. However, in real-world tasks, geometric structures (e.g., object boundaries and elevation jumps) should act as physical constraints on spectral propagation. Therefore, LiDAR geometry is injected into the generation process of the selective SSM parameters.

Given the HSI main branch feature $X' \in \mathbb{R}^{L\times D_{inner}}$ and the LiDAR token sequence $X_{geo} \in \mathbb{R}^{L\times D}$, a geometric prior is first extracted from LiDAR features. To capture local elevation discontinuities, a simple finite-difference operator is used to obtain the local gradient of LiDAR tokens. The geometric prior is formulated as follows:

$$G = \phi_g\left(\left[X_{geo}, \nabla X_{geo}\right]\right) \tag{7}$$

where ∇X_{geo} is the local geometric gradient, $[\cdot,\cdot]$ is the channel-wise concatenation, and $\phi_g(\cdot)$ is the composed of a linear projection, a depth-wise 1-D convolution, and a linear mapping. The depth-wise convolution extracts local geometric contexts with low computational cost. Meanwhile, the HSI

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branch generates the basic selective SSM parameters as follows:

$$\delta^h, B^h, C^h = Split\left(Linear\left(X'\right)\right) \tag{8}$$

where $\delta^h \in \mathbb{R}^{L\times R}$ is the low-rank representation of the discretization step, $B^h, C^h \in \mathbb{R}^{L\times N}$ are the basic input and output projection parameters, R is the low-rank dimension, and N is the state dimension. To establish cross-modal parameter interaction, the HSI feature, the LiDAR geometric prior, and their multiplicative interaction are jointly considered

$$\Theta = \phi_m\left(\left[X', G, X' \odot G\right]\right) \tag{9}$$

where $\phi_m(\cdot)$ is a lightweight multilayer perceptron (MLP) and $\odot$ denotes element-wise multiplication. The term $X'\odot G$ explicitly models the local consistency between spectral semantics and geometric structures, which distinguishes the modulation from simple feature concatenation or linear summation.

The modulation factors are obtained by splitting Θ along the channel dimension as follows:

$$\tilde{s}^\Delta, \tilde{r}^\Delta, \tilde{s}^B, \tilde{r}^B, \tilde{s}^C, \tilde{r}^C = Split\left(\Theta\right) \tag{10}$$

where $\tilde{s}^\Delta, \tilde{r}^\Delta \in \mathbb{R}^{L\times R}$ and $\tilde{s}^B, \tilde{r}^B, \tilde{s}^C, \tilde{r}^C \in \mathbb{R}^{L\times N}$. Here, s and r denote the multiplicative scale factor and residual factor, respectively. To avoid excessive perturbation of the original SSM parameters, these modulation factors are bounded by the hyperbolic tangent function

$$s^m = \epsilon \tanh\left(\tilde{s}^m\right) \qquad r^m = \epsilon \tanh\left(\tilde{r}^m\right) \tag{11}$$

where $m \in \{\Delta, B, C\}$, ϵ is a small constant controlling the modulation range. Finally, the geometry-modulated selective parameters are calculated as follows:

$$\begin{aligned} \delta &= \left(1 + s^\Delta\right) \odot \delta^h + r^\Delta \\ B &= \left(1 + s^B\right) \odot B^h + r^B \\ C &= \left(1 + s^C\right) \odot C^h + r^C \\ \Delta &= Softplus\left(Linear_{dt}\left(\delta\right)\right) \end{aligned} \tag{12}$$

where Δ, B, and C are the final selective SSM parameters used in the scan process. LiDAR geometry adaptively regulates the discretization step, input injection, and state readout of the HSI sequence. This mechanism enables Geo-ΔSSM to perform parameter-level cross-modal interaction.

3) Selective Scan and Gating: With the geometry-modulated parameters Δ, B, and C, the continuous system parameters are discretized using the zero-order hold (ZOH) rule for $\bar{A}$ and $\bar{B}$. The selective scan operation is performed recursively to model global spectral dependencies as follows:

$$\begin{aligned} \bar{A} &= \exp\left(\Delta A\right) \qquad \bar{B} = \left(\Delta A\right)^{-1}\left(\exp\left(\Delta A\right) - I\right) \cdot \Delta B \\ h_t &= \bar{A} h_{t-1} + \bar{B} x_t \qquad y_t = C_t h_t \end{aligned} \tag{13}$$

where $t \in \{1, \ldots, L\}$ is the token index, h_t is the hidden state, and A is a structured learnable parameter. The matrix A is not directly modulated by LiDAR data to preserve the stability of the recurrent state transition. The LiDAR-guided modulation is imposed on Δ, B, and C, which, respectively, control the discretization step, input injection, and state readout. Finally, the output is modulated by the gating branch Z using a multiplicative gate as $Y_{out} = Linear(y \odot SiLU(Z))$, ensuring that only relevant spectral features are propagated.

4) Block Aggregation: After the GSPM process, CME incorporates residual connections and a dynamic MLP to facilitate deep network training. Specifically, the lth block is formulated as follows:

$$\begin{aligned} X'_l &= CrossMamba\left(LN\left(X_{l-1}\right), X_{geo}\right) + X_{l-1} \\ X_l &= MLP\left(LN\left(X'_l\right)\right) + X'_l \end{aligned} \tag{14}$$

where LN represents layer normalization, $CrossMamba(\cdot)$ is the aforementioned procedures in CME, and X_{l-1} and X_l are the input and output of the lth block. The MLP layer further refines the features to ensure a robust representation for downstream classification.

5) Global Context Aggregation and Classification: After deep feature abstraction by the CME blocks, the output sequence tokens X_l are aggregated along the sequence dimension using a GAP operation. Subsequently, a linear classifier head is employed to map the aggregated representations into the final class probabilities, which are utilized for optimization.

F. Dual-Modal Boundary-Aware Loss

In order to solve the problem that boundaries are difficult to classify, a DBAL is presented for Geo-ΔSSM, which utilizes HSI and LiDAR data to identify material boundaries and elevation boundaries, respectively. Since HSI contains numerous spectral bands and significant noise, a spectral intensity map is considered to reduce its dimensionality. The Sobel operator is then applied to compute the gradient G as follows:

$$\begin{aligned} \bar{X}_H\left(i,j\right) &= \frac{1}{bn}\sum_{k=1}^{bn} X_H\left(i,j,k\right) \\ G_M\left(i,j\right) &= \sqrt{\left(S_x * I\right)^2 + \left(S_y * I\right)^2}, \quad I \in \left\{X_L, \bar{X}_H\right\} \end{aligned} \tag{15}$$

where $*$ is the convolution operation, bn represents the number of HSI bands, S_x and S_y represent the horizontal and vertical Sobel kernels, $M \in \{L, H\}$. After obtaining the gradient, α and β are used to adjust the weights over geometric boundaries and material boundaries as follows:

$$w\left(i,j\right) = 1 + \alpha \cdot \frac{G_L\left(i,j\right)}{Max\left(G_L\right) + \epsilon} + \beta \cdot \frac{G_H\left(i,j\right)}{Max\left(G_H\right) + \epsilon} \tag{16}$$

where ϵ is a fixed small number. A larger weight $w(i, j)$ enables Geo-ΔSSM to impose a greater penalty on more salient boundaries, while $w(i, j) = 1$ ensures effective regularization in flat regions. Furthermore, considering the severe long-tail distribution in the datasets, a class-balancing term is introduced to penalize the suppression of minority classes. Notably, N_c denotes the number of training samples for the cth category. The smoothed class weight is initially calculated as $1/(N_c)^{1/2}$. To maintain the overall magnitude of the loss and stabilize gradient optimization, these weights are subsequently normalized by their mean across all classes, denoted as λ_c. By integrating the spatial boundary weight $w(i, j)$ and the semantic class weight λ_c, the combined supervision forces the model to focus on both hard-to-classify boundaries and rare objects. Finally, the loss function is formulated as follows:

$$L_{total} = -\frac{1}{N}\sum_{n=1}^{N}\left(w_n \lambda_{y_n} \sum_{k=1}^{K} y_{n,k} \log\left(p_{n,k}\right)\right) \tag{17}$$

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Algorithm 1 Overall Workflow of Geo-ΔSSM

Require: HSI patch X_{in}^{H}, LiDAR patch X_{in}^{L}, label Y
Ensure: Prediction logits p

Extract shallow HSI feature X^H by Conv3D and HetConv.
Extract LiDAR feature X^L by stacked Conv2D layers.
Generate LiDAR-guided frequency mask: $\gamma = \text{Sig}(\text{Linear}(GAP(X^L)))$, $M = W_{\text{static}} \odot \gamma$.
Enhance X^H by FrEM: $\hat{X}^H \leftarrow \text{IFFT}(\text{FFT}(X^H) \odot M) + X^H$.
Flatten $\hat{X}^H$ and X^L into HSI tokens X_{hsi} and LiDAR tokens X_{geo}.
Add positional embeddings to X_{hsi}.
for each CME block **do**
    Generate main branch feature X' and gating branch feature Z from X_{hsi}.
    Generate HSI-based selective parameters δ^h, B^h, and C^h from X'.
    Compute LiDAR geometry representation: $G = \phi_g([X_{\text{geo}}, \nabla X_{\text{geo}}])$.
    Model cross-modal parameter interaction: $\Theta = \phi_m([X', G, X' \odot G])$.
    Generate bounded modulation terms: $(\tilde{s}^{\Delta}, \tilde{r}^{\Delta}, \tilde{s}^{B}, \tilde{r}^{B}, \tilde{s}^{C}, \tilde{r}^{C}) = \text{Split}(\Theta)$.
    Bound the scale and residual terms: $s^m = \epsilon \tanh(\tilde{s}^m), r^m = \epsilon \tanh(\tilde{r}^m), m \in \{\Delta, B, C\}$.
    Modulate selective parameters: $\delta = (1 + s^{\Delta}) \odot \delta^h + r^{\Delta}, B = (1 + s^{B}) \odot B^h + r^{B}, C = (1 + s^{C}) \odot C^h + r^{C}$.
    Obtain $\Delta = \text{Softplus}(\text{Linear}_{dt}(\delta))$.
    Update X_{hsi} by geometry-conditioned selective scan with Δ, B, and C.
    Apply gating branch and output projection: $X_{\text{hsi}} \leftarrow \text{Linear}(Y \odot \text{SiLU}(Z))$.
    Refine X_{hsi} with an MLP layer.
end for
Obtain prediction logits p by mean pooling and a linear classifier.
Compute the dual-modal boundary-aware classification loss during training.
return p

where N is the batch size, K is the number of classes, and $y_{n,k}$ and $p_{n,k}$ are the true label and predicted probability of the nth sample for the kth class, respectively. $w_n = w(i_n, j_n)$ represents the spatial boundary weight assigned to the nth sample, where (i_n, j_n) is the center spatial coordinate of the input patch in the original data.

IV. Experimental Results and Analysis

A. Dataset Description

The Trento dataset was acquired over a rural area south of Trento, Italy, combining HSI from the AISA Eagle sensor and LiDAR-derived DSM from the Optech ALTM 3100EA sensor. There are 600 × 166 pixels with a spatial resolution of 1 m. The HSI contains 63 spectral bands ranging from 402.89 to 989.09 nm. Table I provides information on the training and testing samples for each class. The HSI and DSM visualization is shown in Fig. 4.

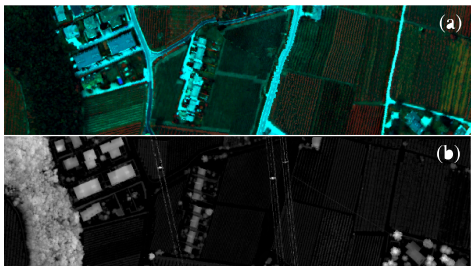


Fig. 4. Visualization of the Trento dataset, including (a) pseudo-color HSI and (b) gray image of the LiDAR DSM data.



Fig. 5. Visualization of the Houston dataset, including (a) pseudo-color HSI and (b) gray image of the LiDAR DSM data.

TABLE I
Training and Testing Sample Numbers in the Trento Dataset

Class	Category	Training	Testing	Total
1	Apple Trees	41	3993	4034
2	Buildings	30	2873	2903
3	Ground	5	474	479
4	Wood	92	9031	9123
5	Vineyard	106	10395	10501
6	Roads	32	3142	3174

TABLE II
Training and Testing Sample Numbers in the Houston Dataset

Class	Land-cover Category	Training	Testing	Total
1	Healthy Grass	13	1238	1251
2	Stressed Grass	13	1241	1254
3	Synthetic Grass	7	690	697
4	Tree	13	1231	1244
5	Soil	13	1229	1242
6	Water	4	321	325
7	Residential	13	1255	1268
8	Commercial	13	1231	1244
9	Road	13	1239	1252
10	Highway	13	1214	1227
11	Railway	13	1222	1235
12	Parking-lot1	13	1220	1233
13	Parking-lot2	5	464	469
14	Tennis-court	5	423	428
15	Running-track	7	653	660

The Houston dataset was captured over the urban environment of the University of Houston campus and its surroundings for the IEEE GRSS Data Fusion Contest in 2013. Its HSI, collected by the compact airborne spectrographic imager (CASI), consists of 144 spectral channels ranging from 380 to 1050 nm with a spatial resolution of 2.5 m. The scene covers a 349 × 1905 pixel grid and includes a co-registered DSM derived from LiDAR data. The sample numbers in all 15 categories are provided in Table II. The HSI and DSM visualization is shown in Fig. 5.

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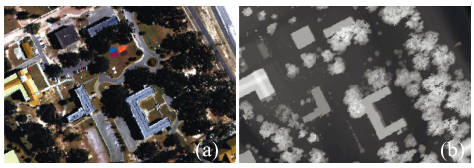


Fig. 6. Visualization of the MUUFL dataset, including (a) pseudo-color HSI and (b) gray image of the LiDAR DSM data.

TABLE III
TRAINING AND TESTING SAMPLE NUMBERS IN THE MUUFL DATASET

Class	Land-cover Category	Training	Testing	Total
1	Trees	233	23013	23246
2	Mostly Grass	43	4227	4270
3	Mixed Ground Surface	69	6813	6882
4	Dirt and Sand	19	1807	1826
5	Road	67	6620	6687
6	Water	5	461	466
7	Building Shadow	23	2210	2233
8	Building	63	6177	6240
9	Sidewalk	14	1371	1385
10	Yellow Curb	2	181	183
11	Cloth Panels	3	266	269

The MUUFL dataset was collected over the University of Southern Mississippi Gulf Park campus. This dataset provides high-resolution (1 m) HSI and LiDAR-derived DSM, covering a spatial extent of 325 $\times$ 220 pixels. The HSI acquired by the CASI-1500 sensor originally contained 72 bands. However, the initial and final bands were removed due to noise artifacts, leaving 64 usable spectral channels. Table III represents the 11 different categories with their training and testing sample numbers. The HSI and DSM visualization is shown in Fig. 6.

To evaluate the classification performance in few-shot scenarios, the training set is constructed by randomly selecting 1% samples from each category across the three datasets. The remaining 99% samples are then used for testing.

B. Experimental Setup

All experiments were implemented with PyTorch on a workstation equipped with three NVIDIA GeForce RTX 2080 Ti GPUs and an Intel Core i9-9900X CPU. The adaptive moment estimation (Adam) optimizer was chosen for optimization. The geometric boundary weight α and material boundary weight β were both set to 0.5. The learning rate was set to 0.001. The batch size was selected within {4, 8, 16, 32, 64, 128, 256, 512}. The training epoch number was set to 500. The patch size was chosen within {3, 5, 7, 9, 11, P13}. Five evaluation metrics were selected, including accuracy per category, overall accuracy (OA), average accuracy (AA), Kappa coefficient (κ), and $F1$ score ($F1$). All experiments were repeated five times to ensure statistical reliability. The means and standard deviations (stds) of the results are reported. The former mitigates the impact of random sample selection bias, and the latter reflects the robustness and stability. Fifteen state-of-the-art methods are selected for comparison, including three classic fusion paradigms (early, middle, and late fusion) [3], EndNet [3], standard CNN-based model *CNNH* [5], multiview hierarchical network (MVHN) [36], self-calibrated convolution-based multiscale deep learning network (MSNetSC) [37], local information interaction transformer (LIIT) [38], morphological convolution and attention calibration network (MCAC) [39], spatial–spectral salient reinforcement network (Sal^{2}RN) [40], hashing-based deep metric learning (HDML) [41], hybrid lightweight Mamba (HLMamba) [42], light-weighted fusion vision Mamba (LWSCFV) [43], spatial–spectral Mamba (S^{2}Mamba) [44], and dual-feature attention hybrid GCN Mamba (DAHGM) [45]. All networks were fine-tuned to their optimal configurations. Their details are as follows.

1) *EndNet* utilizes fully connected layers for input reconstruction without considering spatial dependencies.
2) *CNN$_H$* considers stacked CNN layers followed by classification heads for HSI feature mapping.
3) *MVHN* extracts multimodal features by Gabor filtering and principal component analysis, employing a specific stacking-based residual module for fusion.
4) *MSNetSC* integrates hierarchical residual learning with self-calibrated convolutions and self-attention for superior feature discrimination.
5) *LIIT* captures local dependencies through a transformer-based multihead self-attention mechanism.
6) *MCAC* utilizes morphological convolutions alongside attention calibration for effective joint classification.
7) *Sal^{2}RN* aligns cross-layer interaction and center spectrum correction to emphasize spatial–spectral information.
8) *HDML* learns discriminative fusion representations by leveraging attention-motivated modules to capture complex intermodal correlations.
9) *HLMamba* integrates CNN-based encoding, Mamba-based fusion, and wavelet-enhanced decoding to capture intramodal, intermodal, and global–local dependencies. Different from HLMamba, our Geo-ΔSSM focuses on HSI-LiDAR classification and injects LiDAR-derived geometric priors into the selective SSM parameter generation process.
10) *LWSCFV* employs skip-sampling spatial fusion VMamba and frequency-domain channel fusion to achieve efficient multimodal remote sensing classification. Different from LWSCFV, our method does not rely mainly on lightweight scanning or frequency-domain channel mixing, but performs geometry-conditioned modulation of selective parameters for adaptive HSI-LiDAR interaction.
11) *S^{2}Mamba* leverages spatial–spectral SSMs to extract contextual information with linear complexity, utilizing a mixture gate to dynamically integrate features. Different from *S^{2}Mamba*, our Geo-ΔSSM explicitly exploits LiDAR geometry to modulate the selective SSM parameters rather than only fusing learned spatial–spectral representations at the feature level.
12) *DAHGM* captures complementary local and global context through hybrid GCN and Mamba networks, enhanced by an attention mechanism for robustness. Different from DAHGM, our method emphasizes geometry-guided parameter-level interaction by using

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TABLE IV

CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), κ ($\times$100), AND $F1$ ($\times$100), ON THE TRENTO DATASET

Class	Early	Middle	Late	EndNet	CNN_H	MVHN	MSNetSC	LIIT	MCAC	Sal^{2}RN	HDML	HLMamba	LWSCFV	S^{2}Mamba	DAHGM	Our
1	94.82±3.25	88.73±5.17	94.84±1.77	89.90±3.12	97.43±0.57	92.57±12.61	75.19±36.85	97.07±1.44	99.58±0.27	77.10±38.86	99.28±0.91	99.04±0.26	**99.70±0.24**	97.94±0.79	98.97±0.28	99.16±0.41
2	93.45±1.38	90.63±5.83	96.88±0.36	92.23±1.80	81.24±8.15	98.42±0.48	94.50±3.34	98.38±0.84	99.51±0.26	93.64±8.85	94.02±3.65	97.40±1.37	99.45±0.44	**99.92±0.05**	99.16±0.40	98.29±0.03
3	44.01±15.39	61.01±26.99	33.71±4.19	47.05±12.92	98.52±0.58	91.73±10.92	80.46±17.22	96.79±3.86	67.22±9.58	18.40±28.76	78.95±19.73	87.89±5.91	90.89±11.75	97.59±1.69	98.99±0.99	**100.00±0.00**
4	99.95±0.03	99.59±0.34	99.99±0.01	99.97±0.04	99.31±0.46	99.98±0.03	98.74±1.90	99.98±0.03	99.99±0.01	99.59±0.46	99.57±0.32	99.96±0.05	99.95±0.05	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
5	96.90±1.65	97.10±1.57	98.39±0.45	97.98±1.42	99.86±0.08	99.92±0.02	93.00±12.63	95.12±8.37	99.21±1.33	55.77±37.83	99.35±1.01	99.26±0.42	**99.99±0.01**	99.37±0.19	99.97±0.02	99.93±0.07
6	84.65±1.87	91.67±3.60	85.37±1.96	95.18±1.30	94.49±2.49	94.36±1.14	95.72±2.38	90.66±2.54	94.91±0.66	92.51±5.12	93.51±1.52	95.34±0.65	92.84±2.32	93.13±0.53	95.41±0.72	**98.14±0.62**
OA	95.09±0.39	94.97±0.65	95.86±0.22	95.85±0.45	96.99±0.56	98.10±1.90	92.59±6.17	96.72±2.71	98.57±0.49	78.75±15.62	97.96±1.01	98.67±0.30	98.99±0.24	98.74±0.18	99.27±0.06	**99.50±0.14**
AA	85.63±2.23	88.12±4.75	84.86±0.60	87.05±2.16	95.14±1.08	96.16±3.97	89.60±8.65	96.33±1.01	93.40±1.65	72.83±11.24	94.11±4.02	96.48±1.13	97.14±1.92	97.99±0.30	98.75±0.17	**99.25±0.19**
κ	93.45±0.51	93.28±0.88	94.47±0.29	94.45±0.59	95.99±0.74	97.44±2.57	90.07±8.35	95.68±3.52	98.09±0.65	73.57±18.82	97.28±1.34	98.23±0.39	98.66±0.32	98.31±0.24	99.03±0.08	**99.33±0.19**
F1	86.44±2.00	88.89±3.75	86.29±0.71	89.12±2.15	94.17±1.13	96.38±2.75	88.53±9.18	93.88±1.87	95.09±1.33	67.70±15.50	94.23±4.04	97.18±0.82	97.26±1.29	97.62±0.45	98.39±0.07	**98.71±0.47**

TABLE V

CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), κ ($\times$100), AND $F1$ ($\times$100), ON THE HOUSTON DATASET

Class	Early	Middle	Late	EndNet	CNN_H	MVHN	MSNetSC	LIIT	MCAC	Sal^{2}RN	HDML	HLMamba	LWSCFV	S^{2}Mamba	DAHGM	Our
1	85.70±0.00	76.54±14.18	93.59±4.09	86.67±3.90	82.21±13.69	97.29±0.85	81.26±25.87	61.21±27.21	79.24±31.79	70.23±28.38	97.51±0.32	93.09±3.04	96.98±1.01	95.93±1.47	94.62±4.59	**97.72±0.60**
2	81.64±8.04	91.65±5.44	55.29±37.67	94.13±3.42	97.03±4.85	89.85±0.00	85.19±8.25	89.11±11.21	81.74±13.35	72.09±25.46	86.04±2.91	92.65±5.70	88.07±1.44	89.19±0.37	91.09±4.69	**97.73±0.90**
3	90.72±2.07	82.35±5.06	95.45±2.54	89.77±3.20	86.52±8.48	91.30±1.93	84.20±12.05	97.07±3.10	96.43±2.56	73.42±36.76	85.74±2.24	95.13±2.81	98.70±0.64	93.83±3.11	97.36±3.11	**99.30±0.36**
4	89.37±2.70	74.17±9.93	65.41±15.33	80.63±6.65	93.66±3.88	98.91±0.52	97.32±0.95	83.75±8.16	92.10±7.41	87.86±11.28	95.06±1.29	91.73±7.21	**99.81±0.20**	98.55±0.41	99.59±0.31	99.79±0.24
5	97.23±3.81	88.92±5.54	76.03±26.11	96.99±1.01	85.16±12.44	99.25±0.25	73.00±37.25	97.30±3.68	94.14±11.11	97.53±3.52	96.99±0.88	98.88±1.03	98.96±1.53	99.93±0.06	99.30±0.79	**100.00±0.00**
6	73.15±1.78	41.81±4.72	60.81±4.52	49.84±2.89	63.18±19.59	88.47±0.81	74.83±9.38	90.47±6.31	58.57±13.59	17.26±21.00	71.46±6.15	65.79±4.97	85.36±2.98	79.50±2.62	88.60±3.83	**95.08±1.19**
7	77.24±3.92	33.48±4.60	36.76±6.93	51.84±2.92	62.60±26.04	81.71±1.86	66.20±16.78	78.55±7.65	73.98±5.79	56.18±22.67	78.65±4.90	67.46±12.39	**88.24±3.10**	82.45±2.41	87.70±2.69	85.93±0.92
8	72.20±3.68	77.71±6.19	83.40±3.06	82.49±5.73	70.01±3.41	77.24±6.58	73.97±7.72	67.44±11.00	87.78±6.90	71.28±6.15	73.60±9.03	79.74±4.47	78.98±8.89	80.94±2.37	**88.40±4.16**	80.65±6.65
9	65.57±21.98	37.87±10.38	23.42±3.59	49.91±3.26	81.10±7.78	**82.68±3.59**	50.57±28.79	49.48±16.95	63.54±7.95	42.87±23.65	57.42±2.90	52.77±22.52	72.07±7.28	78.64±2.46	77.09±4.51	79.64±3.94
10	44.66±16.19	59.41±3.51	60.03±25.24	68.17±3.35	44.33±12.25	79.44±5.16	53.92±20.05	62.32±21.93	63.33±11.58	58.12±16.98	76.16±8.33	76.57±15.28	76.87±11.52	68.40±6.05	76.13±17.40	**95.62±2.13**
11	47.94±4.72	32.06±13.39	18.63±11.10	30.36±3.68	92.05±6.28	81.55±3.93	42.42±21.76	39.48±23.17	58.71±22.58	76.86±8.99	86.46±5.27	80.31±13.80	91.15±2.46	84.09±4.50	86.76±6.32	**93.11±3.92**
12	43.67±25.20	66.62±3.77	53.51±17.07	68.80±1.89	18.43±21.47	82.46±3.21	43.69±27.65	71.59±19.63	83.11±12.46	49.62±14.16	79.25±7.23	73.59±4.06	**93.18±1.32**	80.48±3.80	90.85±5.83	88.33±1.20
13	16.64±5.52	10.43±4.52	14.44±6.00	10.99±3.57	43.62±26.12	67.24±9.23	50.56±27.03	64.83±18.82	25.39±11.56	8.66±13.69	87.59±4.52	33.41±13.72	88.88±7.03	86.08±2.74	91.16±5.07	**95.26±0.86**
14	16.69±8.15	38.91±9.66	24.87±22.40	48.09±1.14	97.02±1.82	98.72±0.95	28.56±36.54	81.42±12.78	90.83±11.67	67.80±21.97	96.83±2.07	86.24±8.29	92.34±12.19	99.24±1.19	99.86±0.28	**99.95±0.09**
15	98.84±0.66	78.90±7.88	85.76±5.25	82.14±3.27	99.57±0.53	99.88±0.11	57.49±32.34	95.93±3.87	93.84±6.42	92.68±7.03	94.00±4.49	97.83±2.55	**100.00±0.00**	96.54±2.52	99.88±0.25	**100.00±0.00**
OA	69.63±2.34	62.49±1.78	57.55±6.91	69.39±1.12	74.13±6.47	87.55±0.77	65.80±13.61	73.05±6.06	77.67±7.57	66.59±7.71	83.66±1.61	80.45±2.76	89.47±1.67	86.97±0.66	90.37±2.34	**92.95±0.51**
AA	66.75±1.95	59.39±1.82	56.49±6.29	66.06±1.28	74.43±7.35	87.73±0.93	64.21±13.58	75.33±6.05	76.18±6.83	62.83±7.98	84.18±1.42	79.01±1.63	89.97±2.04	87.59±0.68	91.23±2.25	**93.87±0.46**
κ	67.11±2.53	59.37±1.93	54.02±7.53	66.82±1.22	71.99±7.03	86.54±0.83	63.04±14.60	70.89±6.55	75.88±8.10	63.75±8.40	82.32±1.74	78.83±2.98	88.62±1.81	85.92±0.71	89.59±2.53	**92.38±0.56**
F1	66.05±2.38	59.08±1.76	55.50±5.93	66.75±1.11	74.80±7.86	88.51±0.85	62.88±15.53	74.13±6.48	76.08±8.43	62.40±8.51	84.94±1.50	79.99±2.07	90.45±1.86	87.50±0.97	91.66±2.09	**93.87±0.45**

TABLE VI

CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), κ ($\times$100), AND $F1$ ($\times$100), ON THE MUUFL DATASET

Class	Early	Middle	Late	EndNet	CNN_H	MVHN	MSNetSC	LIIT	MCAC	Sal^{2}RN	HDML	HLMamba	LWSCFV	S^{2}Mamba	DAHGM	Our
1	95.04±0.84	94.61±0.88	95.04±1.42	91.59±2.89	92.53±6.86	96.70±0.32	93.47±2.96	95.32±0.83	**97.48±0.26**	93.95±2.19	96.36±0.58	95.86±0.63	96.62±0.68	97.35±0.26	97.45±0.23	95.65±0.18
2	51.96±18.86	69.57±10.75	57.61±17.58	69.08±18.89	74.01±17.57	74.96±5.58	51.65±30.80	70.82±9.91	76.53±7.70	71.66±14.00	74.35±1.50	71.59±8.79	72.79±9.55	70.57±2.24	73.42±4.51	**77.98±2.23**
3	79.40±7.47	65.83±15.08	66.36±19.46	62.09±26.14	72.37±12.84	75.06±5.61	67.13±11.46	63.64±17.98	75.46±8.26	71.98±12.25	73.67±2.70	79.86±5.44	77.13±9.80	74.97±1.94	77.33±4.35	**81.89±1.52**
4	79.61±8.18	64.16±8.16	83.35±4.98	59.24±17.88	82.22±4.15	74.63±6.38	52.50±15.38	62.45±14.35	81.13±3.41	78.73±11.32	71.74±2.90	85.62±3.07	76.97±18.00	76.37±2.02	78.23±6.03	**86.15±2.00**
5	86.15±3.44	89.12±4.95	89.37±2.52	85.94±7.33	88.88±5.66	95.24±1.21	91.23±1.46	90.23±2.38	**95.53±0.68**	83.22±20.13	93.75±0.81	91.94±2.25	94.77±1.68	94.14±0.80	95.39±0.63	93.47±0.39
6	21.69±2.85	65.21±1.67	77.66±3.90	62.91±4.35	77.57±1.54	68.89±4.21	62.69±31.20	53.54±13.37	67.94±4.74	29.89±36.66	70.80±1.97	69.46±2.88	56.57±25.42	69.93±1.42	77.92±5.13	**95.66±2.01**
7	68.72±3.65	62.10±5.75	64.05±3.36	53.11±4.91	74.53±2.32	75.07±4.21	74.24±7.61	76.78±8.33	69.68±7.07	57.55±11.81	71.36±4.04	71.27±6.86	64.23±12.89	65.81±1.34	73.31±2.85	**80.69±0.68**
8	93.68±1.17	93.31±2.08	93.73±1.19	94.90±1.10	89.23±6.66	96.31±0.69	93.93±2.16	92.85±1.99	96.31±0.48	76.32±21.18	96.22±0.48	94.87±1.94	95.71±1.12	96.42±0.34	**97.32±0.58**	94.38±0.41
9	47.92±5.73	41.50±12.30	41.58±3.95	39.29±11.54	57.01±4.83	54.76±5.26	44.35±9.88	58.79±12.63	52.34±3.67	11.73±10.62	50.56±3.49	51.29±8.16	49.48±14.89	55.36±4.04	59.29±5.13	**61.78±2.98**
10	1.99±0.75	2.65±1.18	5.64±1.83	1.88±1.87	5.41±2.33	5.30±0.56	2.32±2.35	3.43±3.45	4.09±1.29	0.00±0.00	4.53±0.81	4.75±4.89	6.19±3.41	7.18±1.60	6.08±0.49	**68.51±2.96**
11	29.85±5.85	16.84±5.53	32.86±7.43	22.33±11.70	28.57±3.97	21.58±1.32	26.92±7.71	46.47±9.32	29.47±10.17	6.32±8.52	20.38±8.44	43.53±5.30	52.48±10.42	**65.56±12.55**	60.45±8.22	45.99±0.77
OA	84.23±0.29	83.38±1.60	83.69±1.62	80.75±1.98	84.87±4.98	88.31±0.35	82.17±4.30	84.64±2.15	88.82±0.43	80.51±3.25	87.39±0.43	87.95±0.44	87.78±1.39	88.03±0.27	89.38±0.46	**89.90±0.10**
AA	59.64±0.66	60.45±2.03	64.29±1.68	58.40±3.62	67.49±2.31	67.14±1.34	60.04±5.78	64.94±2.33	67.81±1.49	52.85±6.15	65.79±0.87	69.09±1.61	67.54±6.35	70.33±0.81	72.38±1.70	**80.20±0.28**
κ	79.06±0.37	77.93±2.20	78.28±2.30	74.66±2.60	80.24±6.01	84.49±0.48	76.53±5.52	79.72±2.76	85.12±0.57	74.04±4.42	83.25±0.55	84.04±0.62	83.73±1.95	84.07±0.35	85.87±0.61	**86.71±0.14**
F1	60.59±0.44	61.85±2.64	64.22±1.76	59.02±3.91	68.52±3.68	69.44±0.77	59.94±7.47	65.43±3.54	70.90±1.43	52.16±6.64	68.53±1.23	71.09±0.96	70.23±4.93	71.54±0.80	74.77±1.18	**76.88±0.45**

LiDAR-derived structural features to guide the generation of Δ, B, and C in the selective SSM.

C. Experimental Results and Comparison

Experiments are conducted on these three datasets to validate the effectiveness of Geo-ΔSSM. The quantitative classification accuracies are recorded in Tables IV–VI. The superior accuracies are highlighted in bold. In general, classic pixel-wise and CNN-based approaches (e.g., CNN_H and MVHN) yield suboptimal performance due to their limited capacity for modeling long-range dependencies and complex spatial contexts. Furthermore, cross-modal techniques consistently outperform single-modal baselines, verifying that the fusion of HSI spectral details and LiDAR geometric structures is feasible and beneficial for distinguishing complex objects.

Among all compared methods, Geo-ΔSSM achieves the highest performance across all four evaluation metrics on all three datasets. It is worth noting that although OA reflects global correctness, AA serves as a more critical indicator for assessing the unbiased performance, particularly in regions with severe long-tail distributions. As shown in Table VI, although Geo-ΔSSM improves OA by 0.52% over the second-best method DAHGM on the MUUFL dataset, the improvement in AA reaches 7.82%, further enhancing its superiority. Similarly, on the Houston and Trento datasets, Geo-ΔSSM outperforms the second-best methods and improves AA by 2.64% and 0.50%, respectively. Such substantial improvements in AA demonstrate that Geo-ΔSSM does not merely prioritize majority classes to increase OA, but also effectively captures discriminative features for minority categories that are underrepresented and poorly handled by standard fusion networks.

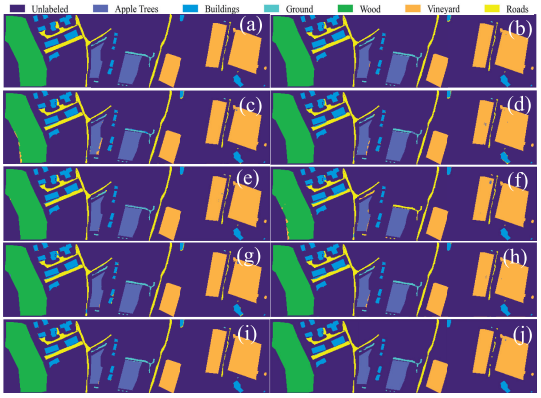


Fig. 7. (a) Ground truth and (b)–(j) classification maps on the Trento dataset, including (b) MVHN, (c) MSNetSC, (d) LIIT, (e) MCAC, (f) Sal^{2}RN, (g) HDML, (h) S^{2}Mamba, (i) DAHGM, and (j) Geo-ΔSSM.

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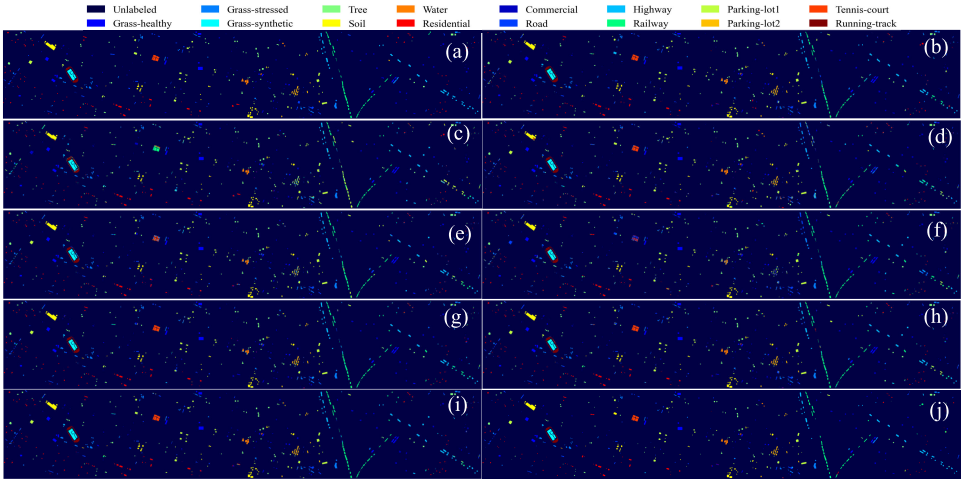


Fig. 8. (a) Ground truth and (b)–(j) classification maps on the Houston dataset, including (b) MVHN, (c) MSNetSC, (d) LIIT, (e) MCAC, (f) Sal^{2}RN, (g) HDML, (h) S^{2}Mamba, (i) DAHGM, and (j) Geo-ΔSSM.

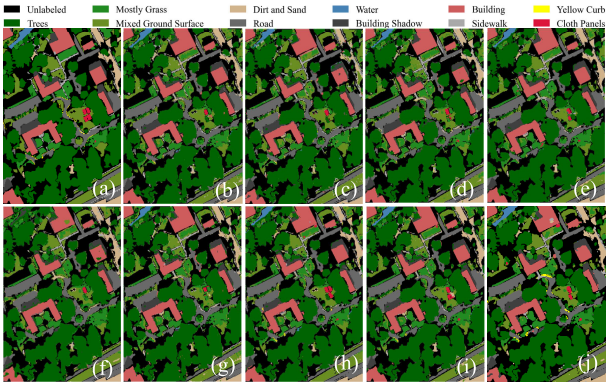


Fig. 9. (a) Ground truth and (b)–(j) classification maps on the MUUFL dataset, including (b) MVHN, (c) MSNetSC, (d) LIIT, (e) MCAC, (f) Sal^{2}RN, (g) HDML, (h) S^{2}Mamba, (i) DAHGM, and (j) Geo-ΔSSM.

This balanced performance is primarily attributed to the proposed GSPM strategy and the DBAL. In categories with limited training samples, such as *building shadow* (Class 7) and *cloth panels* (Class 11) in the MUUFL dataset, classic approaches struggle to extract robust features, resulting in low identification rates. However, Geo-ΔSSM utilizes LiDAR-derived geometric priors to constrain spectral feature propagation, effectively regularizing the learning process for these tail classes. Consequently, Geo-ΔSSM achieves 80.69% accuracy on the challenging *building shadow* category, outperforming DAHGM by 7.38%. The superior κs and $F1$ scores confirm that the prediction consistency and precision-recall balance of Geo-ΔSSM are robust against class imbalance.

The classification maps in Figs. 7–9 further validate these quantitative results. The maps generated by Geo-ΔSSM exhibit high homogeneity within regions and sharp delineation at boundaries. For instance, the *mixed ground surface* samples in the MUUFL dataset can be identified with significantly less salt-and-pepper noise by Geo-ΔSSM compared to other approaches. These qualitative results prove that Geo-ΔSSM successfully mitigates the interference of complex environmental factors while preserving fine-grained details for both large- and small-scale objects.

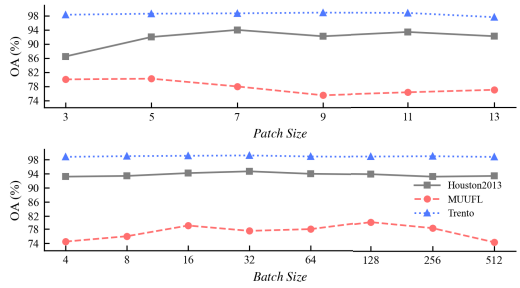


Fig. 10. Classification performance regarding OA (%) using different patch sizes and batch sizes on the three datasets.

D. Parameter Selection

1) Patch Size Selection: The classification accuracies for different patch sizes are illustrated in Fig. 10. In general, larger patch sizes provide more spatial information for feature extraction. However, excessively large patch sizes introduce irrelevant neighboring samples, leading to degraded performance. For the Houston dataset, AA improves as the patch size increases to 7, but drops by 2.33% when the size reaches 13, suggesting that a moderate window of 7 effectively captures urban structural geometry without over-smoothing. The best performance is obtained when the patch size is 5 on the MUUFL dataset. As the window size increases, accuracy declines substantially, indicating that larger windows introduce redundant information in such fragmented scenes. In contrast, the highest accuracy is attained when the patch size is 9 on the Trento dataset, showing that a broader region is beneficial for homogeneous areas such as vineyards. Thus, the patch sizes are set to 7, 5, and 9 for the three datasets, respectively.

2) Batch Size Selection: The performances on the three datasets are also shown in Fig. 10. For the Houston and Trento datasets, the optimal performance is achieved when the batch size is 32. The MUUFL dataset requires a larger batch size of 128 to obtain satisfactory performance. This result suggests that larger batch sizes help stabilize optimization in scenarios with severe class imbalance and noise. Thus, the optimal batch sizes are set to 32, 128, and 32 for the respective datasets.

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TABLE VII

CLASSIFICATION PERFORMANCE WITH RESPECT TO AA (%) USING DIFFERENT α AND β VALUES ON THE MUUFL DATASET

α \ β	0	0.25	0.5	0.75	1
0	78.57	78.63	78.73	78.21	78.34
0.25	77.59	77.53	77.52	77.97	77.70
0.5	77.14	77.03	**79.14**	78.00	78.49
0.75	77.81	78.06	78.32	77.44	77.19
1	77.52	78.84	78.35	77.04	78.08

TABLE VIII

CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), κ ($\times$100), AND $F1$ ($\times$100), WITH DIFFERENT VALUES OF R

R	MUUFL dataset				Trento dataset				Houston dataset			
	OA	AA	Kappa	F1	OA	AA	Kappa	F1	OA	AA	Kappa	F1
1	89.72	77.93	86.45	76.57	99.28	98.78	99.04	98.25	92.51	93.59	91.90	93.44
2	89.66	77.87	86.37	75.89	99.23	98.73	98.98	98.11	92.92	**93.96**	92.35	93.85
4	89.81	79.74	86.58	76.51	99.26	98.76	99.01	98.18	92.71	93.69	92.12	93.62
8	89.64	78.58	86.38	76.12	99.25	98.74	98.99	97.95	92.35	93.44	91.73	93.31
16	89.81	76.73	86.59	75.37	99.25	98.55	99.00	98.00	92.45	93.58	91.84	93.51
32	89.55	78.34	86.24	76.64	99.21	98.52	98.95	97.91	92.54	93.60	91.94	93.58
64	89.76	78.42	86.51	76.40	99.18	98.41	98.90	97.77	92.54	93.62	91.94	93.54
auto	**89.90**	**80.20**	**86.71**	**76.88**	**99.50**	**99.25**	**99.33**	**98.71**	**92.95**	93.87	**92.38**	**93.87**

3) Boundary Weight Selection: Boundary weights α and β explicitly correspond to the penalty strengths for LiDAR-derived geometric boundaries and HSI-derived material boundaries in the loss formulation, respectively. To determine their optimal values, grid search experiments were conducted on the three datasets, with candidate values chosen from $\{0, 0.25, 0.5, 0.75, 1\}$. Given the highly consistent performance trends across different scenes, the AA results on the MUUFL dataset are considered in Table VII. The highest accuracy is obtained when both α and β are set to 0.5, achieving an AA of 79.14%. This balanced setting effectively combines the LiDAR elevation jumps and HSI spectral variations, guiding Geo-ΔSSM to preserve fine-grained structural edges without distorting latent feature spaces. Since the Houston and Trento datasets exhibit similar optimal performance under this setting, both α and β are fixed to 0.5.

4) Low-Rank Dimension Selection: The classification accuracies with different values of R are reported in Table VIII. The low-rank dimension R directly controls the capacity for geometry-spectral interaction in GSPM. Besides fixed values of R, we also evaluate an automatic setting, denoted as "auto" in Table VIII, which follows $R = \lceil D/16 \rceil$, where D denotes the hidden dimension. A small R may restrict the expressive ability of the geometry-conditioned modulation, while an excessively large R may introduce redundant parameters and unstable optimization. As shown in Table VIII, the automatic setting achieves the best overall performance on the three datasets. Specifically, it obtains the highest OA, AA, κ, and $F1$ on the MUUFL and Trento datasets. On the Houston dataset, although $R = 2$ achieves a slightly higher AA, the automatic setting obtains the best OA, κ, and $F1$. Therefore, the automatic setting is selected as the final low-rank dimension configuration of Geo-ΔSSM because it strikes a favorable balance between interaction capacity and optimization stability without dataset-specific manual selection.

E. Computational Complexity Analysis

To further evaluate the computational efficiency of the proposed Geo-ΔSSM, we compare different methods in terms of the number of parameters, FLOPs, training time per epoch, inference time per sample, and peak GPU memory consumption. The results on the MUUFL, Trento, and Houston datasets are reported in Table IX.

From the perspective of model size, Geo-ΔSSM maintains a moderate number of parameters, i.e., 0.25, 0.27, and 0.32 M on the MUUFL, Trento, and Houston datasets, respectively. Although it introduces additional geometry-conditioned modulation branches, its parameter scale is much smaller than that of several deep fusion models, such as MVHN, MSNetSC, LIIT, and HDML. This suggests that the proposed selective parameter modulation does not rely on a large number of learnable parameters.

In terms of FLOPs, Geo-ΔSSM requires 13.83-, 47.12-, and 38.71-M FLOPs on the three datasets. Its computational cost is higher than that of some lightweight baselines, such as S^2Mamba and LWSCFV, because Geo-ΔSSM introduces FrEM and GSPM to explicitly model frequency-domain enhancement and geometry-conditioned selective parameter generation. However, compared with methods with heavier feature extraction or fusion modules, such as MVHN, MSNetSC, LIIT, HDML, and HLMamba, Geo-ΔSSM still keeps the computational cost within a practical range.

For runtime efficiency, Geo-ΔSSM shows competitive inference speed. The inference time is 0.01, 0.03, and 0.03 ms/sample on the MUUFL, Trento, and Houston datasets, respectively, which is comparable to or faster than most of the compared methods. Although the training time per epoch is relatively higher due to the frequency-domain processing and GSPM, this additional training cost is acceptable because inference remains efficient after training. Moreover, the peak GPU memory consumption of Geo-ΔSSM is 183.43, 438.93, and 251.05 MB on the three datasets, which is lower than many recent fusion models and demonstrates competitive space efficiency.

Theoretically, let N denote the number of tokens, D denote the hidden dimension, S denote the SSM state dimension, and R denote the low-rank dimension in GSPM. The selective SSM scanning in CME has linear complexity with respect to the token length. More specifically, its computational complexity is approximately $\mathcal{O}(NDS)$, which simplifies to $\mathcal{O}(ND)$ when S is fixed. This avoids the quadratic complexity $\mathcal{O}(N^2D)$ of self-attention.

The proposed GSPM introduces additional geometry-conditioned modulation for the low-rank discretization representation δ and the selective parameters B and C. Its additional cost is approximately $\mathcal{O}(ND(R + 2S))$, which is linear with respect to the token number and controlled by the small low-rank dimension R and state dimension S. The FrEM module further introduces FFT/IFFT-based frequency-domain operations with a cost of approximately $\mathcal{O}(DN \log N)$. In terms of space complexity, the main activation memory of CME is approximately $\mathcal{O}(ND)$, while the additional modulation variables in GSPM require about $\mathcal{O}(N(R + 2S))$ storage.

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TABLE IX
COMPUTATIONAL COMPLEXITY COMPARISON ON THE MUUFL, TRENTO, AND HOUSTON DATASETS

Dataset	Metric	Early	Middle	Late	EndNet	CNN_H	MVHN	MSNetSC	LIIT	MCAC	Sal^{2}RN	HDML	HLMamba	LWSCFV	S^{2}Mamba	DAHGM	Our
MUUFL	Params (M)	0.45	0.12	0.49	0.49	0.04	2.37	0.62	0.37	**0.02**	0.38	1.01	0.24	0.04	0.09	0.34	0.25
	FLOPs (M)	0.90	**0.23**	0.97	0.98	1.85	118.05	31.96	11.19	0.38	15.58	48.20	57.78	0.99	1.54	6.77	13.83
	Train/epoch (s)	**0.03**	0.06	0.04	0.08	0.03	0.06	0.08	0.07	0.07	0.07	0.08	0.12	0.16	0.05	0.14	0.72
	Infer. (ms/sample)	0.01	0.02	0.01	0.02	0.01	0.03	0.04	0.03	0.03	0.05	0.03	0.10	0.02	**0.01**	0.07	0.01
	Memory (MB)	368.03	610.54	363.92	1078.88	541.97	1217.91	580.70	594.70	759.29	1925.93	764.05	537.10	640.94	640.75	**153.73**	183.43
Trento	Params (M)	1.36	0.34	1.40	1.41	0.04	2.37	1.05	0.38	**0.03**	0.37	1.02	0.24	0.05	0.09	0.34	0.27
	FLOPs (M)	2.72	**0.69**	2.79	2.81	5.94	382.37	173.52	35.23	1.19	49.18	151.18	57.52	3.20	5.00	21.90	47.12
	Train/epoch (s)	**0.05**	0.10	0.07	0.14	0.06	0.11	0.15	0.13	0.14	0.13	0.14	0.14	0.30	0.06	0.14	1.10
	Infer. (ms/sample)	0.03	0.04	0.04	0.05	0.03	0.07	0.07	0.06	0.06	0.11	0.07	0.11	0.02	**0.01**	0.09	0.03
	Memory (MB)	696.22	1104.29	673.91	1953.69	843.12	1957.62	1130.91	1062.85	1356.94	3133.85	1350.20	483.69	1120.50	1120.62	**155.97**	438.93
Houston	Params (M)	1.86	0.47	1.89	1.90	0.06	2.42	4.14	1.58	**0.04**	1.92	5.18	0.27	0.05	0.10	0.35	0.32
	FLOPs (M)	3.71	**0.93**	3.78	3.79	5.88	235.89	414.17	110.38	1.00	154.35	476.90	78.43	2.19	3.15	13.51	38.71
	Train/epoch (s)	**0.02**	0.04	0.02	0.05	0.02	0.04	0.06	0.04	0.05	0.05	0.05	0.05	0.09	0.03	0.04	0.50
	Infer. (ms/sample)	0.05	0.06	0.05	0.08	0.05	0.07	0.09	0.08	0.07	0.15	0.10	0.19	0.02	**0.01**	0.09	0.03
	Memory (MB)	521.87	689.93	497.58	1336.46	930.71	1363.31	870.28	939.73	630.38	2434.46	1284.81	614.76	776.91	776.54	**162.66**	251.05

F. Ablation Study

1) Necessity of Network Modules: To demonstrate the necessity of employing all network modules, diverse alternative structures were constructed by removing or changing partial modules from Geo-ΔSSM, which are listed as follows.

FEM-related structures

1) K_1: removes the whole FEM module from Geo-ΔSSM.
2) K_2: replaces the HetConv2D layers with Conv2D layers.

FrEM-related structures

3) K_3: removes the whole FrEM module from Geo-ΔSSM.
4) K_4: preserves the static branch and removes the dynamic branch in the FrEM module.
5) K_5: preserves the dynamic branch and removes the static branch in the FrEM module.
6) K_6: removes the residual connection in the FrEM module.

CME-related structures

7) K_7: removes the whole CME module from Geo-ΔSSM.
8) K_8: replaces the CME module with classic Mamba blocks.
9) K_9: replaces the proposed GSPM with direct concatenation of HSI and LiDAR tokens before the Mamba block.
10) K_{10}: removes the LiDAR gradient prior ∇X_{geo} in GSPM.
11) K_{11}: removes the multiplicative interaction term $X' \odot G$ in GSPM.
12) K_{12}: only modulates the discretization step size Δ, while keeping B and C generated from the HSI branch.
13) K_{13}: modulates Δ and B, while keeping C generated from the HSI branch.
14) K_{14}: removes the positional embeddings from tokenization steps in Geo-ΔSSM.

Their classification accuracies are presented in Table X. Specifically, removing or changing any designed component tends to degrade classification performance, confirming the indispensability of these modules in Geo-ΔSSM. The accuracies of K_1 and K_2 on the three datasets are consistently lower than those of Geo-ΔSSM, highlighting the essential role of the FEM module. These structures cannot extract sufficient shallow spatial–spectral representations from raw heterogeneous data, inevitably leading to suboptimal performance in the subsequent sequence modeling.

TABLE X
CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), κ ($\times$100), AND $F1$ ($\times$100), TO PROVE THE NECESSITY OF NETWORK MODULES

Ablation Structure	MUUFL dataset				Trento dataset				Houston dataset			
	OA	AA	Kappa	F1	OA	AA	Kappa	F1	OA	AA	Kappa	F1
K_1	87.69	76.71	83.90	71.59	98.66	97.19	98.22	96.25	92.00	93.13	91.35	93.02
K_2	89.50	77.26	86.16	75.79	99.30	98.77	99.07	98.17	90.51	91.72	89.74	91.71
K_3	89.89	78.70	86.70	76.33	98.94	98.20	98.59	97.65	92.43	93.54	91.82	93.43
K_4	89.74	79.20	86.49	76.61	98.94	98.25	98.59	97.74	92.17	93.24	91.54	93.10
K_5	89.67	79.04	86.41	76.75	99.34	98.95	99.12	98.43	92.42	93.33	91.89	93.28
K_6	**90.01**	75.18	86.81	75.41	99.28	98.73	99.04	98.39	91.60	92.67	90.92	92.76
K_7	89.73	78.83	86.49	76.34	98.99	98.28	98.66	97.92	92.60	93.59	92.00	93.50
K_8	89.69	78.84	86.45	76.32	99.29	98.81	99.05	98.18	92.83	93.80	92.25	93.73
K_9	89.51	77.25	86.47	76.62	99.28	98.81	99.04	98.05	92.34	93.06	91.88	93.10
K_{10}	89.89	78.48	86.70	76.41	99.28	98.76	99.03	98.09	92.45	93.51	91.84	93.46
K_{11}	89.73	78.95	86.47	76.01	99.23	98.70	98.98	98.08	92.12	93.23	91.49	93.11
K_{12}	89.73	78.45	86.48	76.49	99.30	98.67	99.07	98.17	92.56	93.59	91.96	93.51
K_{13}	89.67	78.52	86.40	76.41	99.29	98.83	99.06	98.17	92.66	93.66	92.06	93.55
K_{14}	89.75	79.60	86.53	76.89	99.27	98.81	99.02	98.20	92.85	93.80	92.27	93.69
Our	89.90	**80.20**	**86.71**	**76.88**	**99.50**	**99.25**	**99.33**	**98.71**	**92.95**	**93.87**	**92.38**	**93.87**

The structures K_3–K_6 verify the effectiveness of the proposed FrEM module. Removing the whole FrEM module in K_3 degrades the classification performance, especially on the Trento and Houston datasets, indicating that frequency-domain enhancement is beneficial for improving multimodal feature discrimination. The structures K_4 and K_5 further analyze the roles of the static and dynamic branches in FrEM. When only the static branch is preserved (K_4), the model mainly relies on dataset-specific frequency priors and cannot adapt well to local spectral variations. When only the dynamic branch is preserved (K_5), the model can adjust the frequency response according to input-dependent features, but lacks the global stability provided by the static branch. Thus, integrating both branches is necessary to construct an effective hybrid frequency mask. Furthermore, K_6 removes the residual connection in the FrEM module to investigate whether adding the original HSI features after frequency-domain processing is beneficial. It should be noted that FrEM is not designed as a pure denoising module that completely replaces the original HSI features. Instead, it performs residual frequency enhancement, where the frequency-domain branch provides an adaptive correction term, while the identity path preserves the original spectral signatures that may be weakened by frequency filtering. Compared with Geo-ΔSSM, K_6 decreases the AA and $F1$ on MUUFL by 5.02% and 1.47%, respectively, and also reduces the OA/AA/$F1$ on Houston from

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92.95%/93.87%/93.87% to 91.60%/92.67%/92.76%. Although K_6 obtains a slightly higher OA on MUUFL, its obvious AA degradation indicates that removing the residual path harms class-balanced recognition. These results demonstrate that the residual connection does not simply reintroduce noise but helps preserve useful spectral information while the frequency branch suppresses unfavorable spectral disturbances.

The results of the CME-related structures verify the effectiveness of the proposed GSPM. When removing the whole CME module in K_7, the classification performance drops on all three datasets, demonstrating the necessity of geometry-aware long-range dependency modeling and cross-modal interaction. Replacing the proposed CME with classic Mamba blocks in K_8 also leads to inferior results, indicating that directly applying standard SSMs is insufficient for HSI and LiDAR fusion because the selective parameters are mainly generated from the input sequence itself without explicit geometric constraints.

More importantly, K_9 replaces the proposed GSPM with direct concatenation of HSI and LiDAR tokens before the Mamba block, which directly evaluates whether simple feature-level fusion can replace the proposed geometry-conditioned parameter modulation. Compared with K_9, Geo-ΔSSM improves the AA by 2.95%, 0.38%, and 0.81% on the MUUFL, Trento, and Houston datasets, respectively. This demonstrates that simply concatenating LiDAR features to the SSM input cannot fully exploit geometric priors. In contrast, GSPM injects LiDAR-derived geometric features into the generation process of selective parameters, allowing the model to adapt the scanning dynamics.

K_{10} removes the LiDAR gradient prior ∇X_{geo} in GSPM. Its degraded performance confirms that explicit geometric discontinuity information is useful for guiding selective parameter generation. K_{11} removes the multiplicative interaction term $X' \odot G$, which weakens the local geometry feature interaction and leads to consistent performance degradation, especially on the Houston dataset. These results indicate that both the LiDAR gradient prior and the multiplicative interaction term contribute to effective geometry-conditioned modulation.

Furthermore, K_{12} and K_{13} investigate whether modulating only partial selective parameters is sufficient. K_{12} only modulates the discretization step Δ, while keeping B and C generated from the HSI branch. K_{13} modulates Δ and B, while keeping C generated from the HSI branch. Both structures perform worse than the full Geo-ΔSSM, showing that only adjusting the scanning step or the input injection term cannot fully model HSI-LiDAR interactions. The full modulation of Δ, B, and C enables the geometric prior to influence the state transition, input injection, and output projection jointly, resulting in more effective cross-modal unification. Finally, K_{14} removes the positional embeddings from tokenization steps, and its performance drop verifies that explicit positional information is indispensable to compensate for spatial structure loss during sequence flattening.

2) Necessity of Loss Elements: To verify the necessity of loss elements, the structures M_1-M_6 are listed as follows:

1) M_1: uses the standard cross-entropy loss.
2) M_2: replaces the proposed loss with focal loss.
3) M_3: removes the class-balancing weight from the DBAL formulation.
4) M_4: uses the standard inverse class frequency ($\lambda_c = 1/N_c$) as the class weight.
5) M_5: uses the logarithmic inverse frequency ($\lambda_c = 1/ln(N_c)$) as the class weight.
6) M_6: retains the class-balancing weight but removes the spatial boundary weights.

TABLE XI

CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), κ ($\times$100), AND $F1$ ($\times$100), TO PROVE THE NECESSITY OF LOSS ELEMENTS

Ablation Structure	Metrics	Datasets		
		MUUFL	Trento	Houston
M_1	OA	**90.36**	99.43	92.61
	AA	72.27	98.94	93.46
	Kappa	86.22	99.24	92.01
	F1	74.55	**98.80**	93.63
M_2	OA	89.83	99.39	92.24
	AA	70.36	98.99	93.22
	Kappa	86.30	99.19	91.61
	F1	73.21	98.71	93.27
M_3	OA	90.10	99.43	92.49
	AA	69.95	98.98	93.37
	Kappa	86.43	99.24	91.88
	F1	72.44	98.78	93.42
M_4	OA	47.36	98.36	92.50
	AA	69.54	97.90	93.69
	Kappa	43.33	97.82	91.90
	F1	57.47	94.04	93.44
M_5	OA	90.18	99.37	92.68
	AA	76.06	98.87	93.63
	Kappa	86.00	99.16	92.09
	F1	75.52	98.41	93.58
M_6	OA	89.78	99.31	92.57
	AA	78.69	98.91	93.67
	Kappa	86.37	99.09	91.97
	F1	76.55	98.19	93.52
Our	OA	89.90	**99.50**	**92.95**
	AA	**80.20**	**99.25**	**93.87**
	Kappa	**86.71**	**99.33**	**92.38**
	F1	**76.88**	98.71	**93.87**

The classification accuracies of these structures are presented in Table XI. Specifically, the absence or replacement of any designed loss component results in degraded performance, particularly regarding the AA. The results of M_1 and M_2 demonstrate the limitation of conventional loss functions in handling imbalanced remote sensing data. Although M_1 yields a marginally higher OA (90.36%) on the MUUFL dataset than Geo-ΔSSM (89.69%), its AA decreases severely by 6.88% (from 79.15% to 72.27%). This indicates that the standard CE loss biases the network toward majority categories to minimize the global error. The focal loss in M_2 fails to mitigate this phenomenon, further decreasing the AA to 70.36%. The structure M_3 verifies the necessity of the class-balancing mechanism. Removing the class weight causes the AA on the MUUFL dataset to drop to 69.95%, confirming that penalizing spatial boundary errors alone is insufficient for minority class recognition. M_4 and M_5 explore alternative weight calculation strategies. Adopting the standard inverse frequency in M_4 causes a performance collapse on the MUUFL dataset (the OA drops to 47.36%), illustrating that excessive penalties on majority classes destabilize the gradient optimization. While the logarithmic frequency in M_5 limits this extreme distortion, its AA (76.06%) remains lower than that of the adopted smoothed weight. Finally, M_6 removes the spatial boundary weights, leading to degraded metrics across all three datasets (e.g., κ drops from 92.22% to 91.97% on the Houston dataset).

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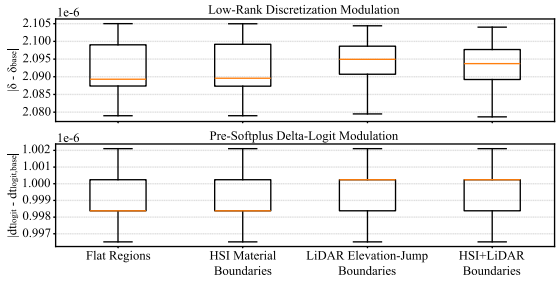


Fig. 11. Boxplots of $|\delta - \delta_{\text{base}}|$ (up) and $|dt_{\text{logit}} - dt_{\text{logit,base}}|$ (down) across different region types.

This decline proves that explicitly penalizing prediction errors near geometric and material transitions is essential for preserving fine-grained structural delineations.

G. Visualization and Statistical Analysis of GSPM

To verify whether the proposed geometry-conditioned modulation behaves as expected, we further analyze the discretization-related variables in the CME block. Specifically, δ_{base} denotes the HSI-only low-rank discretization representation, while δ denotes the geometry-modulated representation. The presoftplus discretization logit is denoted as $dt_{\text{logit}} = \text{Linear}_{dt}(\delta)$, and the final positive discretization step is obtained by $\Delta = \text{Softplus}(dt_{\text{logit}})$. Therefore, the proposed geometry-conditioned modulation affects the generation process of Δ through the intermediate representation δ and the presoftplus discretization logit. Since the final Δ is normalized by the softplus operation and remains numerically stable, we focus on δ and dt_{logit}, which more directly reflect the modulation before positivity normalization.

Testing samples are divided into four region types according to HSI and LiDAR gradient responses, including flat regions, HSI material boundaries, LiDAR elevation jump boundaries, and dual-modal boundaries. We then visualize $|\delta - \delta_{\text{base}}|$ and $|dt_{\text{logit}} - dt_{\text{logit,base}}|$ across these region types.

The low-rank discretization modulation $|\delta - \delta_{\text{base}}|$ exhibits different distributions across region types, as shown in Fig. 11 (up). Compared with flat regions and HSI-only material boundaries, LiDAR elevation-jump boundaries show a slightly higher median modulation response. Similarly, Figs. 11 (down) and 12 show that the presoftplus modulation $|dt_{\text{logit}} - dt_{\text{logit,base}}|$ presents stronger responses in regions containing LiDAR-derived geometric discontinuities. These results suggest that LiDAR-derived geometry modulates the low-rank discretization representation and the presoftplus discretization logits, thereby affecting the generation process of Δ.

This observation is consistent with the structural role of LiDAR-derived DSMs. Elevation-jump boundaries often correspond to structural discontinuities or object edges, where spectral signatures from neighboring materials are more likely to be mixed within local HSI patches. The relatively stronger discretization-related modulation in these regions suggests that the proposed CME is sensitive to geometric discontinuities and may adapt the selective scanning dynamics accordingly, helping constrain spectral information propagation across geometrically inconsistent areas.

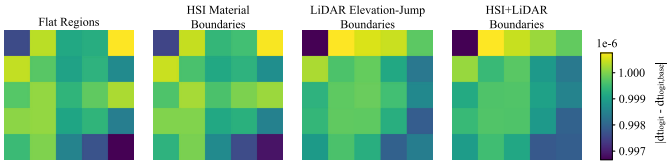


Fig. 12. Averaged heatmaps of presoftplus modulation $|dt_{\text{logit}} - dt_{\text{logit,base}}|$.

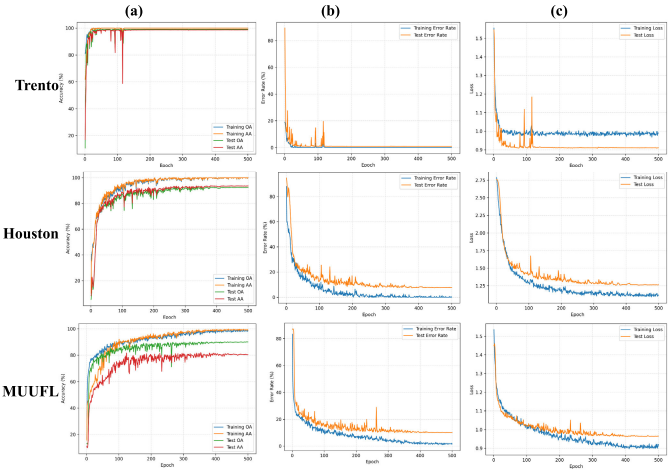


Fig. 13. Convergence analysis of Geo-ΔSSM, (a) training and test accuracy curves, (b) training and test error-rate curves, and (c) training and test loss curves.

H. Convergence and Bias-Variance Analysis

To better understand the convergence behavior and the bias and variance characteristics of Geo-ΔSSM, we further report the accuracy (OA and AA), error rate, and loss curves during training. As shown in Fig. 13, the training loss decreases rapidly in the early stage and then gradually stabilizes. The test loss follows a similar decreasing trend and remains stable in the later epochs, indicating that the model does not show obvious divergence. Meanwhile, the training and test error rates decrease consistently with increasing epochs. The OA and AA curves also increase quickly at the beginning and converge to stable values after sufficient iterations. Although a performance gap exists between the training and test curves due to limited labeled samples and complex class distributions, the gap remains stable after convergence, suggesting that Geo-ΔSSM achieves a reasonable tradeoff between fitting ability and generalization. In addition, the proposed GSPM is designed with bounded perturbations. Specifically, the modulation terms for the selective SSM parameters are constrained by $\epsilon \tanh(\cdot)$, which prevents excessive changes to Δ, B, and C. The final discretization step Δ is further generated through a softplus operation, ensuring positive and numerically stable scanning steps. Therefore, the proposed modulation introduces geometry-aware adaptability while maintaining stable empirical optimization behavior.

V. Conclusion

In this article, a novel geometry-conditioned Δ-modulated selective SSM network called Geo-ΔSSM is proposed to strengthen the coupling capability between HSI and LiDAR

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data and to reduce the computational redundancy in processing high-resolution dense patches. The core motivation is to establish a parameter-level interaction mechanism where LiDAR-derived geometric structures regulate the selective SSM parameters of spectral sequences. This geometric connection helps mitigate the modality gap and improve the physical interpretability of the fusion process. The whole structure consists of three primary components, including FEM, FrEM, and CME. Multiple geometry-aware mechanisms are employed to enhance the feature representation capability in boundary-rich scenarios. Specifically, the FrEM successfully suppresses high-frequency noise by combining static spectral priors with dynamic geometric modulation, while the GSPM allows the model to adaptively adjust Δ, B, and C according to structural continuity. Furthermore, Geo-ΔSSM achieves strong performance in identifying rare classes and delineating sharp boundaries through the guidance of the weighted DBAL. Experiments on three datasets demonstrate that Geo-ΔSSM outperforms other state-of-the-art techniques both qualitatively and quantitatively. Future work will focus on extending this geometry-conditioned modulation paradigm to other multimodal tasks, such as pan-sharpening and change detection.

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